Śabda-ALM: A Speech-Centred Audio Language Model on a Sanskrit Byte-Core, and the Sphoṭa-Lens for Locating Where Sound Becomes Meaning

Pranava Project

2026

Abstract

Text language models begin from already-segmented symbols. Bhartṛhari's *Vākyapadīya* (5th c.), the neuroscience of inner speech, and the frontier of speech language models converge on a different starting point: cognition begins as a continuous acoustic field that bursts into discrete meaning (*sphoṭa*), grasped through an intermediate representation (*madhyamā*/*paśyantī*) before it is uttered (*vaikharī*). We build **Śabda-ALM**, a speech-to-speech Audio Language Model that routes acoustic evidence from a frozen NVIDIA Parakeet encoder, through a trainable *Sphoṭa Projector*, into a from-scratch Sanskrit *byte-core* (a Nemotron-H/Mamba-2 hybrid), and back out through neural TTS. Two results follow. First, on a held-out audio→Sanskrit-text task the Śabda-ALM (CER 0.546) *outperforms* NVIDIA's SOTA Parakeet-TDT native ASR head (0.605, same encoder) and OpenAI Whisper (0.747): routing sound through the Sanskrit core reads Sanskrit speech better than a general decoder. Adapting the frozen core with 1.5M LoRA parameters lifts a 200M model past a 1.13B projector-only model. Second, we introduce the **Sphoṭa-Lens**, an instrument that locates, correlationally and causally, the layer at which the sentence's meaning becomes decodable from the audio positions alone. On the trained model this is *layer 13* of 25 (decodability 0.25 vs 0.022 chance; correlational and causal peaks agree within one layer): sound-borne meaning crystallises mid-network, then moves toward articulation—the *paśyantī*→*vaikharī* gradient made measurable. All results are reproduced by machine-checkable dual-verdict gates.

1 Introduction

Modern language models operate over discrete text tokens—language that has *already* been parsed. The Indian grammarian Bhartṛhari argued that this inverts the order of cognition: awareness is bathed in a continuous field of sound (*dhvani*), and meaning arrives as *sphoṭa*, an indivisible "burst" that is not the sum of its acoustic parts. The tradition organises this into four levels of speech (*vāk*): *parā* (the source), *paśyantī* (the flash of meaning, *pratibhā*), *madhyamā* (mental articulation), and *vaikharī* (uttered sound). Independently, the neuroscience of inner speech and the current frontier of speech language models both move away from discrete text toward continuous acoustic representations.

We take this convergence as an *engineering* directive rather than a metaphor, and ask: can a speech-centred model built on a Sanskrit core (a) read Sanskrit speech better than general SOTA systems, and (b) expose, as a measurable internal locus, the point where sound becomes meaning? We answer both affirmatively, and contribute:

- **Śabda-ALM** (§3): a speech-to-speech ALM built by injecting projected acoustic embeddings into a frozen from-scratch Sanskrit byte-core via its native embedding path, proven faithful (identical predictions to the core's own forward).
- A **SOTA comparison** (§5) in which Śabda-ALM ranks first against Parakeet-TDT and Whisper on the Sanskrit-target task, and a demonstration that LoRA adaptation of the core is a stronger lever than raw parameter scaling.
- The **Sphoṭa-Lens** (§6), a correlational-plus-causal instrument that localises meaning-emergence-from-sound to a validated mid-network layer.

2 Background and Related Work

Sphoṭa and the four *vāk*. Bhartṛhari's *Vākyapadīya* treats the sentence-meaning as *pratibhā*, a distinct cognition arising when word-meanings are grasped together (VP 2.143), not their running sum. We operationalise *paśyantī*/sphoṭa as an internal representation in which meaning is maximally present, and *vaikharī* as the surface (byte) output. **Audio language models.** SALMONN, Qwen2-Audio and NVIDIA's canary-qwen adopt a common recipe: a frozen audio encoder, a trainable projector, and a causal LLM. We follow this recipe but make the LLM a *Sanskrit* byte-core, and add an interpretability instrument. **Global-workspace lenses.** The Jacobian-lens / j-space line locates a text-only workspace band via inter-layer representational similarity. Our Sphoṭa-Lens instead measures, per layer, the *decodability of meaning from the audio positions*, and validates the locus causally by ablation—an audio-native measure the text-only lens cannot express.

3 The Śabda-ALM Architecture

The model realises the four *vāk* as a pipeline:

$$\underbrace{\text{audio}}_{\text{vaikharī in}} \xrightarrow{\text{Parakeet (frozen)}} \underbrace{\text{frames}}_{\text{dhvani}} \xrightarrow{\text{Sphoṭa Projector}} \underbrace{\text{tokens}}_{\text{madhyamā}} \xrightarrow{\text{Sanskrit byte-core}} \underbrace{\text{workspace}}_{\text{paśyantī/sphoṭa}} \to \underbrace{\text{text} \to \text{TTS}}_{\text{vaikharī out}}.$$

Core. A from-scratch 200M-parameter byte-level (vocab = 256) Nemotron-H/Mamba-2 hybrid (d=768, 24 layers, attention every 8) trained on a Sanskrit corpus; and a 1.13B variant (d=1536, 32 layers). The core embeds bytes internally; for audio we bypass the embedding and feed projected acoustic tokens directly into the block stack, adding the core's learned structural-channel bias so the injected tokens live in the distribution the frozen blocks expect. We verify the injection is faithful: feeding embed(ids) through the injection path reproduces the core's own forward (identical arg-max; $< 0.1\%$ logit difference from kernel non-determinism). **Sphoṭa Projector.** A temporal-convolution down-sampler followed by an MLP mapping 1024-dim Parakeet frames to the core's embedding space. **Adaptation.** Beyond a frozen core, we optionally inject rank-r LoRA adapters into the core's MLP projections (the Mamba mixer projections are consumed by a fused kernel and are left intact) and train them jointly with the projector.

4 Experimental Setup

Data. Sentences are realised from Pāṇinian kāraka frames with gold role labels *by construction* (a native derivation engine, replacing a brittle parser dependency), and rendered to audio with neural TTS; 600 paired items, 90/10 train/val by hash. Every item carries its verb (*kriyā*) as a discrete meaning label from the gold parse. **Training.** Frozen encoder and (optionally LoRA-adapted) core; the projector (and LoRA) train under teacher-forced byte cross-entropy. Evaluation is greedy character error rate (CER) on the held-out split, referenced to a no-audio baseline. **Reproducibility.** Each result is guarded by a dual-verdict gate (a machine-checkable `code_gate` and `domain_gate`); heavy compute runs in a pinned NeMo container on an NVIDIA GB10 (200M) and an RTX 5090 (1B).

5 Results

Understanding scales, and adaptation beats scaling. Table 1 shows held-out CER. The no-audio baseline is 1.0 (the core cannot guess the text without audio). Projector-only training already conveys speech content; scaling the core 200M→1.13B lowers CER 0.774 → 0.571; and *LoRA-adapting the* 200*M core* reaches 0.548—below the 1.13B projector-only model with $< 0.1\%$ of its parameters trained.

Configuration	trainable	val CER ↓
200M core, projector-only	projector	0.774
1.13B core, projector-only	projector	0.571
200M core, projector + LoRA(r=8)	projector + 1.5M	**0.548**

Table 1: Held-out character error rate. Adapting the core is a stronger lever than scaling.

SOTA comparison. On the identical held-out audio (Table 2), Śabda-ALM outperforms NVIDIA's Parakeet-TDT native ASR head—*using the same encoder*—and OpenAI Whisper. Routing acoustic evidence through the Sanskrit core reads Sanskrit speech better than a general decoder does. The comparison is fair by construction (same audio, same CER metric) and the *ordering* is the result.

rank	model	CER ↓
1	**Śabda-ALM (Parakeet + Projector + LoRA→Sanskrit core)**	**0.546**
2	Parakeet-TDT-0.6B native ASR (NVIDIA)	0.605
3	Whisper-base (OpenAI)	0.747

Table 2: Held-out audio → romanized-Sanskrit text; lower is better; identical audio for all.

6 The Sphoṭa-Lens: Locating Where Sound Becomes Meaning

We ask, at each layer ℓ, how decodable the sentence's meaning (its kriyā) is from the *audio-token positions alone*. **Correlational**: a template-grouped cross-validated linear probe predicts the kriyā from layer-ℓ audio-position representations; accuracy is the emergence curve. **Causal**: a true activation ablation mean-collapses each layer's audio positions during the forward and measures the drop in final meaning-decodability. A locus is claimed only when the two peaks agree (± 1 layer).

On the trained 200M ALM (240 utterances, 45 kriyā classes, chance = 0.022), meaning is decodable from the audio positions at ≈ 0.25—11× chance. The curve *rises to a peak at layer* 13 (0.263) then declines toward the output (0.25 → 0.23); the correlational peak (13) and causal peak

(14) agree. **The sphoṭa locus is layer 13, validated.** Sound- borne meaning crystallises mid-network, then the representation transforms toward surface form: the *paśyantī*$\to$*vaikharī* gradient, made measurable inside a working model. The instrument is validated on a synthetic core with a planted signal, which it recovers.

7 Discussion

Two threads meet here. As a *product*, a Sanskrit-centred speech model beats general SOTA ASR on Sanskrit, and cheap adaptation of a small core is competitive with a much larger one—an efficient path for low-resource, structure-rich languages. As *interpretability*, the Sphoṭa-Lens turns a classical claim about cognition (meaning bursts from sound at an intermediate level) into a validated statement about a network's internals. The two reinforce each other: the lens tells us *where* the model forms meaning, which is where adaptation and steering should act.

8 Conclusions and Future Work

Śabda-ALM is a working speech-to-speech Audio Language Model on a Sanskrit byte-core that ranks first against SOTA ASR on its target, and the Sphoṭa-Lens localises meaning-emergence-from-sound to a validated mid-network layer. Next: LoRA adaptation of the 1.13B core; native-pronunciation Sanskrit speech; larger corpora; and an autoresearch loop that optimises directly against the live SOTA leaderboard.